

Homework (Habanero)**Q1.** What is the expansion of $(a + b)^2$?

- (A) $a^2 + 2ab + b^2$
 (B) $a^2 - 2ab + b^2$
 (C) $a^2 + ab + b^2$
 (D) $a^2 - ab + b^2$
 (E) $a^2 + b^2$

(A) $a^2 + 2ab + b^2$ (B) $a^2 - 2ab + b^2$ (C) $a^2 + ab + b^2$ (D) $a^2 - ab + b^2$ (E) $a^2 + b^2$ **Q2.** A doctor prescribes a dosage of medicine that is $2x + 5$ mg, where x is the number of days. If the patient takes the medicine for 3 days, what is the total dosage in mg?

- (A) 20 mg
 (B) 21 mg
 (C) 23 mg
 (D) 25 mg
 (E) 30 mg

Q6. A medical researcher collects data on the number of patients who respond to a new treatment. The results are modeled by the equation $y = x^2 - 4x + 4$, where y is the number of patients and x is the dosage level. What is the value of y when $x = 2$?

- (A) 0
 (B) 2
 (C) 4
 (D) 6
 (E) 8

Q3. What is the factored form of $x^2 + 10x + 25$?

- (A) $(x + 5)^2$
 (B) $(x - 5)^2$
 (C) $(x + 10)^2$
 (D) $(x - 10)^2$
 (E) $x(x + 25)$

Q7. The concentration of a solution is given by the equation $C = 2x^2 + 8x + 6$, where C is the concentration and x is the amount of solute. What is the concentration when $x = 1$?

- (A) 12
 (B) 14
 (C) 16
 (D) 18
 (E) 20

Q4. The growth rate of a cell culture is modeled by the equation $P = 2t^2 + 12t + 10$, where P is the population size and t is time in hours. What is the population size after 2 hours?

- (A) 42
 (B) 50
 (C) 52
 (D) 54
 (E) 60

Q8. What is the factored form of $x^2 - 6x + 9$?

- (A) $(x + 3)^2$
 (B) $(x - 3)^2$
 (C) $(x + 6)^2$
 (D) $(x - 6)^2$
 (E) $x(x - 9)$

Q5. What is the expanded form of $(a - b)^2$?**Open Ended Questions (FRQ)****Q9.** A biologist is studying the growth rate of a cell culture. The growth rate is modeled by the equation $P = x^2 + 8x + 10$, where P is the population size and x is time in hours. Find the population size after 3 hours and explain your reasoning. Show all work.**Q10.** A doctor prescribes a dosage of medicine that is $x^2 - 4x + 4$ mg, where x is the number of days. Factor the expression and find the total dosage in mg if the patient takes the medicine for 4 days. Show all work.**Chef's Tips**

- Tip 1: Check for common factors before factoring.
- Tip 2: Use the FOIL method for expanding binomials.
- Tip 3: Verify your answers by plugging in values for variables.